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Which transit hub can Thailand not afford to lose?

Notebook: colab.research.google.com/drive/1WOo4I_1XuAkZx0GXonI8g9ysdZ5dzNx5

1. The decision

The product manager of a Thai tour operator selling Latin America and Caribbean trips decides which transit hubs need a backup route. If all markets had 3+ hubs, they would just buy on price.

2. Why a network

A table counts routes per hub; it cannot say which markets are lost when one hub goes. It depends on whether another hub reaches it too. That is a network-overlap question, not a simple route count.

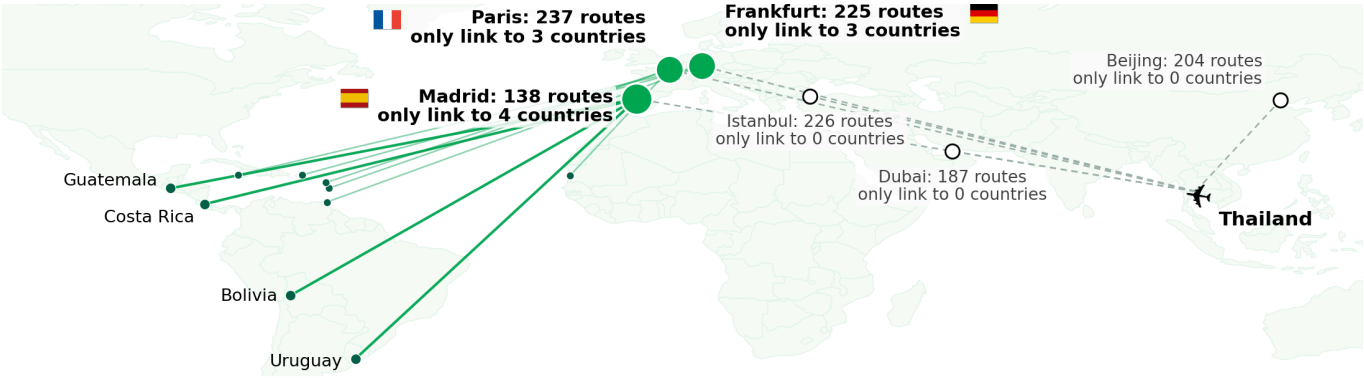
3. Data

Source	OpenFlights routes.dat + airports.dat (github.com/jpatokal/openflights)
Legal basis	ODbL open licence, credit given · verified on public schedules · no personal data
Period · volume	June 2014 · 67,663 route records worldwide
2026 check	Google Flights, 3 Oct 2026 · all 8 only-link hubs still fly nonstop from Bangkok
Limitation	No frequency or seat data · old data, some routes may be gone

4. Graph construction

Scope	Thailand to the 106 countries that need one stop, via 116 airports abroad
Node · Edge	Airport · an airline flies nonstop A to B, directed (codeshares not counted)
Weight · pruning	Unweighted: we ask only if a link exists, and the data has no frequency · left out: Thailand, nonstop countries (Japan, China…), <10 routes
Choice vs alternative	Country vs airport level: by airport, Istanbul (34) and Manila (28) lead, but 28 of each are in their own country; by country, both fall to 0
Robustness	Madrid/Paris/Frankfurt at cutoff 5: 4/5/3 · 10: 4/3/3 · 20: 3/0/3 · codeshares: 4/3/3
Final size	184 nodes, 954 edges, cut to the 15 countries with exactly one hub

5. The graph



Top 3 of 8 such hubs shown. In total 15 countries depend on one hub; 11 are in Latin America and the Caribbean. Data: OpenFlights (ODbL), June 2014.

6. Findings

- **Big is not the same as needed.** Istanbul (226 routes), Beijing (204) and Dubai (187) are the only link to 0 countries; Madrid (138) to 4. Size vs dependency: rank correlation 0.35.
- **The risk sits in one region.** 11 of the 15 one-hub countries are in Latin America and the Caribbean.
- **The result is stable.** Madrid stays in the top two in all four tests; Paris drops out at cutoff 20.

7. Recommended action

- Product manager sets a backup route for Bolivia, Costa Rica, Guatemala and Uruguay by 30 Nov 2026.
- Do not spend on backup for Istanbul, Dubai or Beijing. Their markets all have another hub.
- Watch the Bangkok to Madrid flight. Only one airline flies it; if it stops, the four markets have no one-stop route.

8. What would change my mind

A link is not capacity: the graph shows no seats, frequency or transfer time, and second-leg routes are 2014 data. If the airline quotes a two-stop route that is just as fast, Madrid is not a real dependency.

In short. We mapped which hub links Thailand to each country: the hubs we depend on are not the biggest ones. Backing up Madrid alone protects four markets, so money goes where the risk is.